Moffat Bay Marina

Website Design Document

Blue Group:
Robert Bruetzmann
Miguel Fernandez
Carolina Rodriguez
Sara White

CSD-460
August 16, 2026



Introduction

This document will provide technical implementation details for a website that is to be created for Moffat Bay Marina. Within this document you will find a Terminology section for all stakeholders to reference.

Purpose

Moffat Bay Marina currently offers long-term slip reservations that can be booked in person, over the phone and via email. Their office manager tracks reservations with a spreadsheet that they manually update. Moffat Bay Marina does not currently have a website, so customers must call or email to request information regarding the size of slips and reservation dates available. This process requires careful coordination and manual data entry, which can be time-consuming.

Moffat Bay Marina would like a website that can provide details about the marina and slip availability to any visitors. They would like their website to manage reservations and user accounts for their customers. It is important to note that, while a customer account needs to be created to utilize the online booking system, the website itself should be viewable to any visitors regardless of customer account creation. Additionally, the reservation system will not process payments for customers and will not require payment to confirm a reservation.

Terminology

Marina/Business terms

Slip - a designated area in a marina where a boat can "park" (Discover Boating, n.d.)

Harbormaster - a person who oversees harbor or marina traffic and manages anchorages (Discover Boating, n.d.)

VHF - stands for "Very High Frequency" and is a bandwidth designation typically used by marine radios (Discover Boating, n.d.)

Long-Term Reservation - reservations that last for a season (several months) to an entire year, as opposed to short-term reservations, which are measured in days, weeks or months

Technical/System terms

Authentication - "in web security, it is the process of verifying the claimed identity of some entity, such as a user. This then makes it possible to decide whether to grant the user the access that they are requesting, such as being signed into a particular account." (MDN contributors, 2025a)

User - in most cases, this will refer to the customer, but can refer to any person who is using a system (in this case, the website)

User account – in most cases this will be refer to a customer account, but Moffat Bay Marina staff will have user accounts, as well, with varying permissions.

CustomerID - a unique identifier for each customer to be stored in the Moffat Bay Marina database

ReservationID - a unique identifier for each reservation made to be stored in the Moffat Bay Marina database

Session - a window of time in which a user accesses a website

Landing Page - the main page of a website through which a user can access other sections of the website. The landing page will typically contain basic information about the business and navigation elements, like a menu

Form / Form Field - in HTML, a form contains elements, like fields, that can receive user input (ex. Email field) or user interactions (ex. Clicking the "submit" button after filling out an email field) (MDN contributors, 2025c)

Database - a system that collects and stores organized data. Databases make searching and structuring information easier and more efficient. (Erickson, 2024)

Open Source - publicly available code/designs

RDBMS - stands for Relational Database Management System (Erickson, 2024)

MySQL - an open-source RDBMS that is commonly used to manage databases for companies (Erickson, 2024)

Regular Expression (Regex) - a pattern-matching mechanism that can be used to check whether text conforms to a specific format. For example, a phone number is 10 digits and is typically formatted XXX-XXX-XXXX

ERD - stands for "Entity-Relationship Diagram"; this is a visual flowchart typically used in database design that illustrates how entities (ex. Customers, slip reservations) are related within a system

HTML - stands for "HyperText Markup Language"; this is the standard language that is used to create website content (MDN contributors, 2025c)

CSS - stands for "Cascading Style Sheets"; used for the visual organization of HTML (MDN contributors, 2025c)

JSP - stands for "JavaServer Pages"; a technology that is used to create web pages that can display changing information (dynamic) from a Java-based web application. For example, if a

customer visits the Moffat Bay Marina website and logs into their account, the banner on the webpage might display their name in the upper right corner to indicate they are logged in. (MDN contributors, 2025b)

PII - stands for "Personally Identifiable Information"; this can be any data that can be used to identify a specific person. For example, a person's social security number would be considered PII.

User Personas

User 1: Angela

Gender: Female

Age: 42

Occupation: Elementary Teacher

Education: College/Undergraduate

Marital Status: Married

Location: Local, lives in town year-round

Figure 1



Note. AI-generated image created with Google Gemini (Google, 2026).

Background

Angela and her family have had a 40 ft boat for six years now, and they are looking to reserve a slip in the Moffat Bay area. She's got a full-time job and two kids with their own schedules, so she doesn't have time to sit on the phone dealing with reservations. She just wants something quick that works.

Angela's Needs

- An option to see what slips are open without calling the business
- Some kind of confirmation she can send to her husband once it's booked
- An account that remembers her info so she's not typing the same stuff in every season

Angela's User Stories

1. As a busy teacher, I want to check slip availability online so I can plan my season without calling the office.
2. As a busy parent, I want my account and boat info saved after I register so I don't have to re-enter them every time.
3. As a slip holder, I want confirmation that I can save or forward the reservation details to share with my family.

User 2: Walt

Gender: Male

Age: 68

Occupation: Retired Electrician

Education: High School

Marital Status: Widowed

Location: Winter Visitor

Figure 2



Note. AI-generated image created with ChatGPT (OpenAI, 2026).

Background

Walt is a 68-year-old retired electrician and widower who wants to spend his winter vacation at Moffat Bay. From November through February he plans to stay at Moffat Bay Lodge for three to four months and keep his boat at Moffat Bay Marina so he can spend his days fishing. Since losing his wife, Walt values reliability, simplicity, and a sense of community.

Walt's Needs

- A reserved slip for his entire 3-4 month winter stay
- Easy, clear booking so he can coordinate his stay at the lodge with his marina reservation
- Peace of mind that his boat is secure between fishing trips

Walt's User Stories

1. As a winter dock guest, I want to reserve a dock slip for a full 3–4-month block in one transaction, so I don't have to re-book or risk losing my spot month to month.
2. As a guest who is older, I would prefer to select a dock slip that is closer to the shore, so I don't have to haul my equipment as far when I take it to and from the lodge.
3. As a long-term slip holder, I would like clear, upfront pricing for a multi-month stay, including any seasonal or long-term discounts, so that I can budget for the winter without surprise fees.

User 3: Jesse

Gender: Male

Age: 47

Occupation: Harbormaster

Education: Community College/Vocational School

Marital Status: Married

Location: Local, lives in town year-round

Figure 3



Note. From *Harbor and Marina Staff Directory, by City of Monterey (n.d.).*

Background

This is Jesse, he is a staff member at the Moffat Bay Marina. Jesse's goal is to keep long-term slip reservations organized and make sure customers are assigned to slips that fit their boats.

Jesse's Needs

- Accurate reservations
- Slip availability
- Efficient access to registered customer information
- An easy way to confirm reservations.

Jesse's User Stories

1. As a marina staff member, I want to view current slip availability so that I can know which long-term slips are still available.
2. As a marina staff member, I want customers to be matched with the right slip size based on the length of their boat.
3. As a marina staff member, I want to see confirmed reservations so that I can track available slips and avoid double bookings.

User Story Prioritization

High Priority

1. As a busy teacher, I want to check slip availability online so I can plan my season without calling the office.
2. As a winter dock guest, I want to reserve a dock slip for a full 3–4-month block in one transaction, so I don’t have to re-book or risk losing my spot month to month.
3. As a marina staff member, I want customers to be matched with the right slip size based on the length of their boat.

Medium Priority

4. As a marina staff member, I want to view current slip availability so that I can know which long-term slips are still available.
5. As a marina staff member, I want to see confirmed reservations so that I can track available slips and avoid double bookings.
6. As a long-term slip holder, I would like clear, upfront pricing for a multi-month stay, including any seasonal or long-term discounts, so that I can budget for the winter without surprise fees.
7. As a guest who is older, I would prefer to select a dock slip that is closer to the shore, so I don’t have to haul my equipment as far when I take it to and from the lodge.

Low Priority

8. As a busy parent, I want my account and boat info saved after I register so I don't have to re-enter them every time.
9. As a slip holder, I want confirmation that I can save or forward the reservation details to share with my family.

Development Estimates

The following development estimates break down the three highest priority user stories into individual tasks. Estimated hours are approximate and were developed using the original work estimations as a baseline. Some tasks support multiple user stories, and shared development work is accounted for in the broader project task list.

Story 1: *As a busy teacher, I want to check slip availability online so I can plan my season without calling the office.*

Development Steps	**Hours**
Create database queries to retrieve available slips	2
Implement backend logic to check availability	2
Display available slips on the reservation page	2
Test availability results	1

Total hours: 7

Story 2: As *a winter dock guest, I want to reserve a dock slip for a full 3–4-month block in one transaction, so I don't have to re-book or risk losing my spot month to month.*

Development Steps	**Hours**
Create reservation form	2
Implement reservation processing and database storage	3
Display reservation confirmation	2
Test reservation functionality	1

Total hours: 8

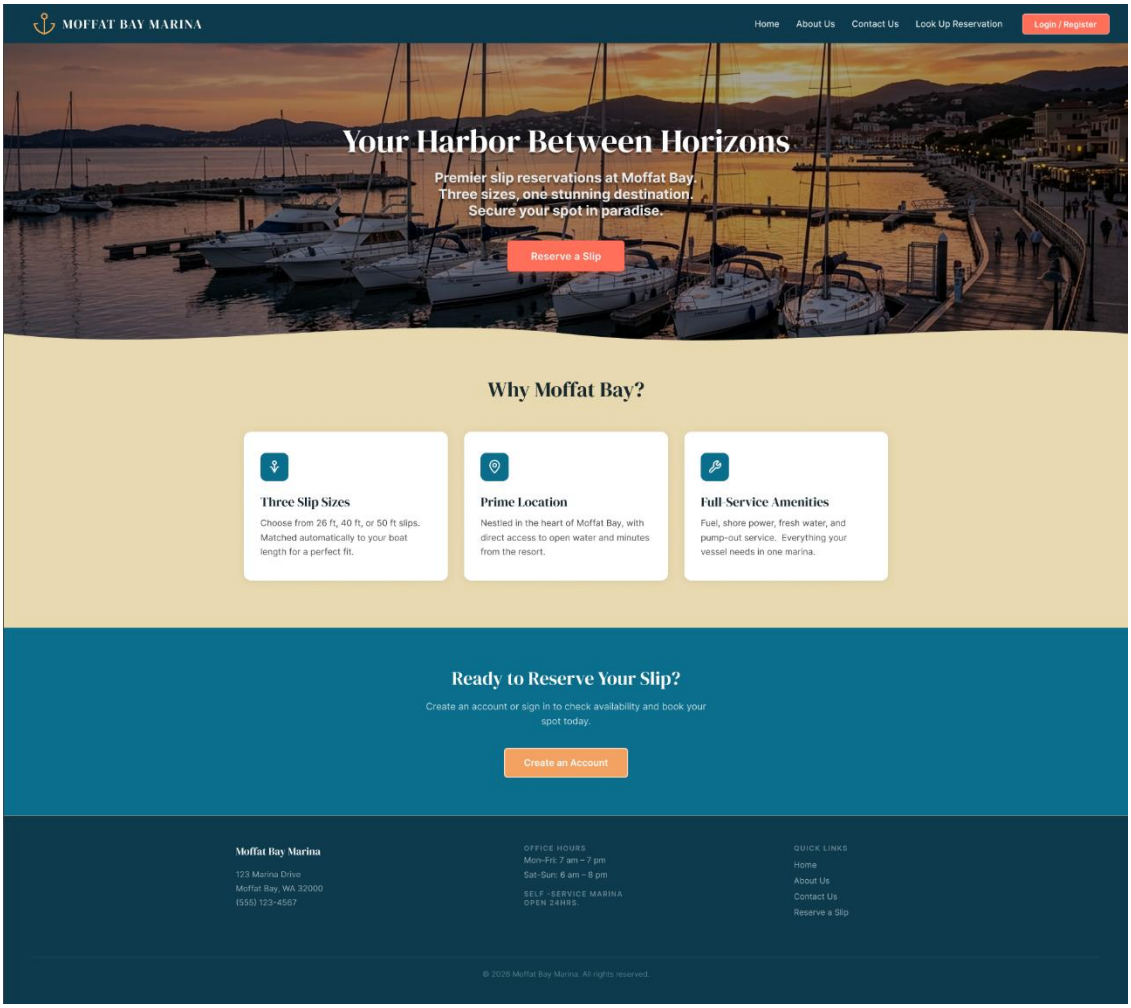
Story *3*: *As a marina staff member, I want customers to be matched with the right slip size based on the length of their boat.*

Development Steps	**Hours**
Define boat-length and slip-size requirements	1
Implement matching logic	2
Integrate matching into the reservation process	1
Test matching and invalid boat lengths	1

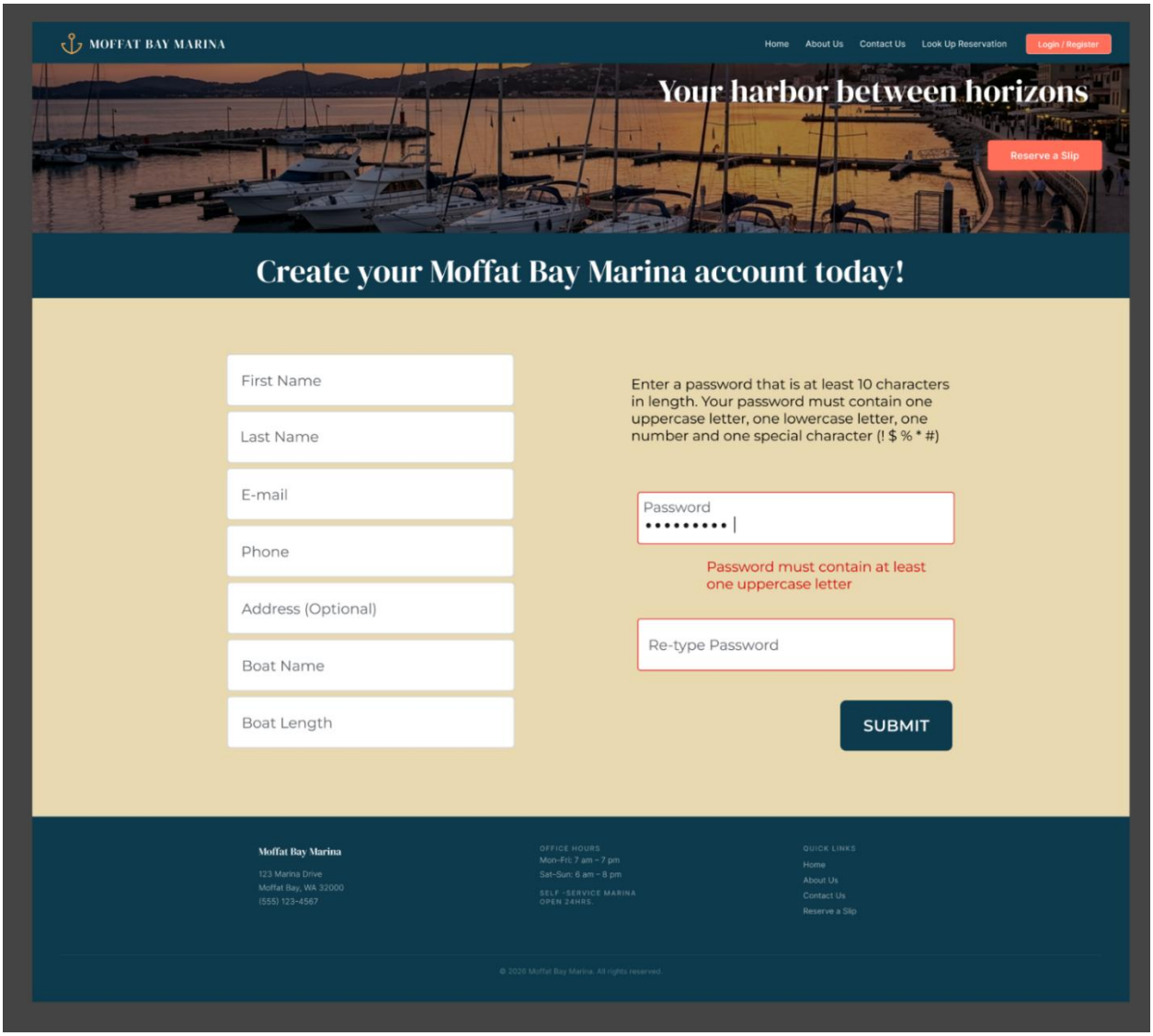
Total hours: 5

Prototypes

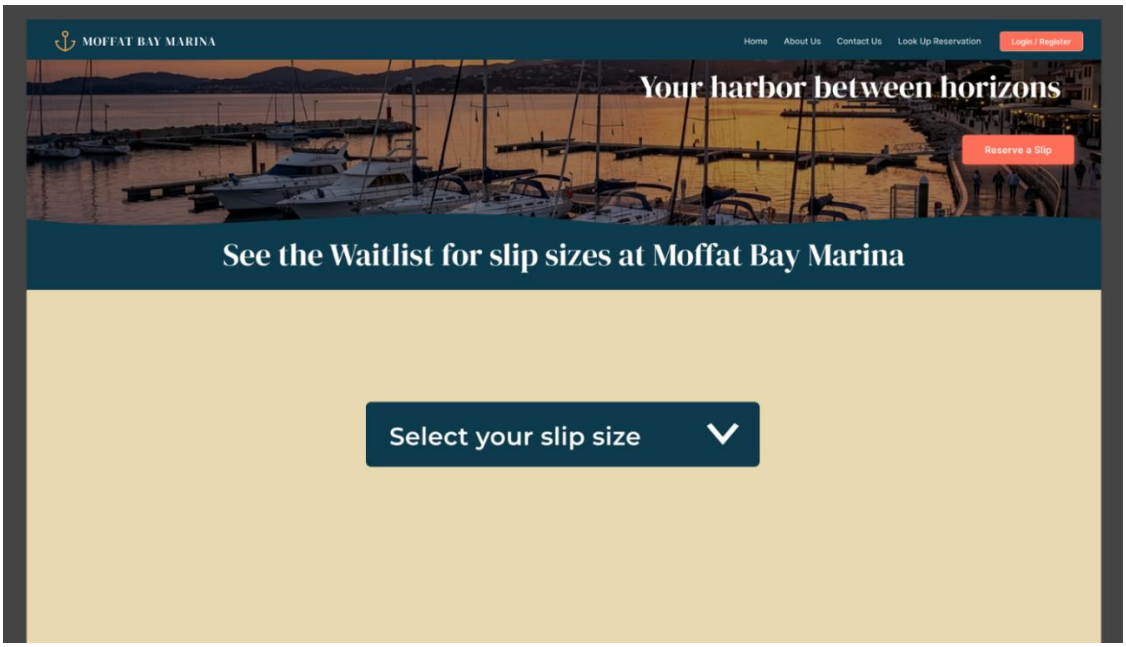
Landing Page:

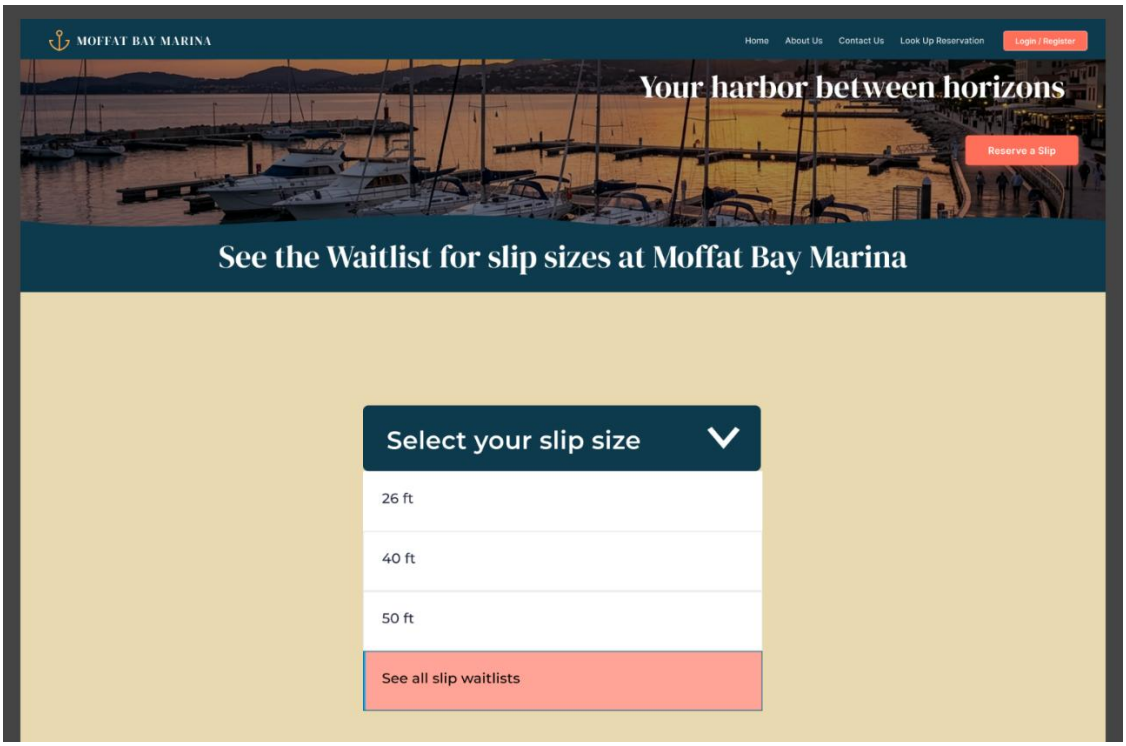


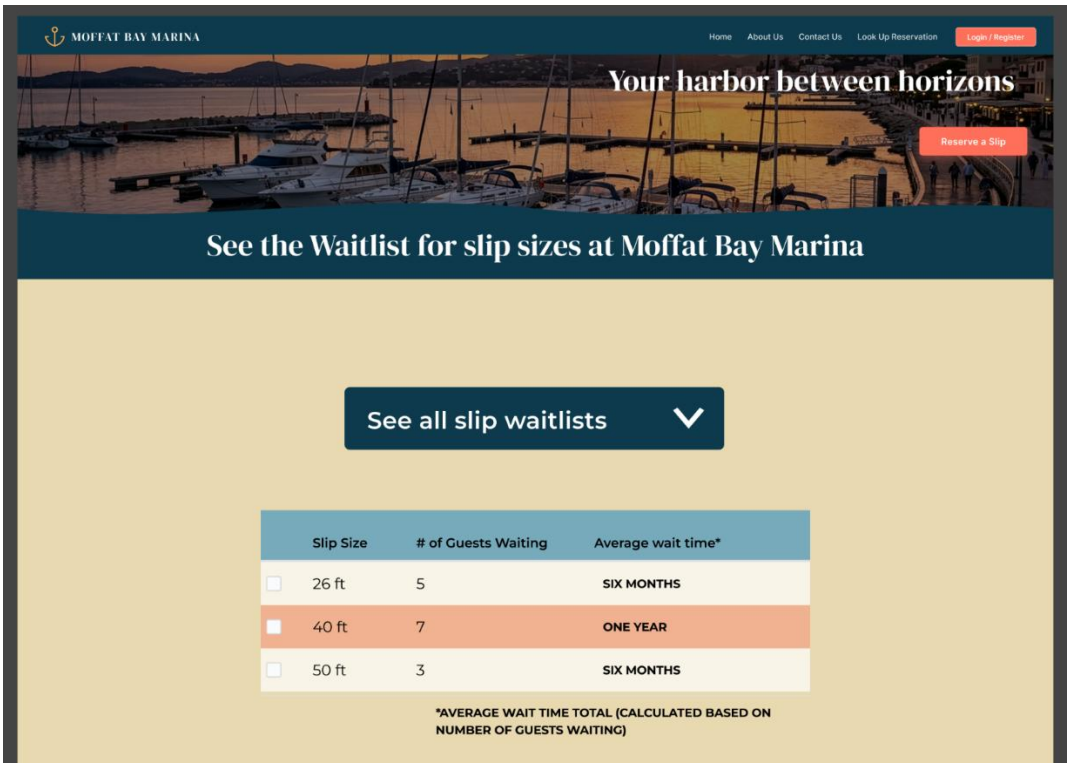
Registration:

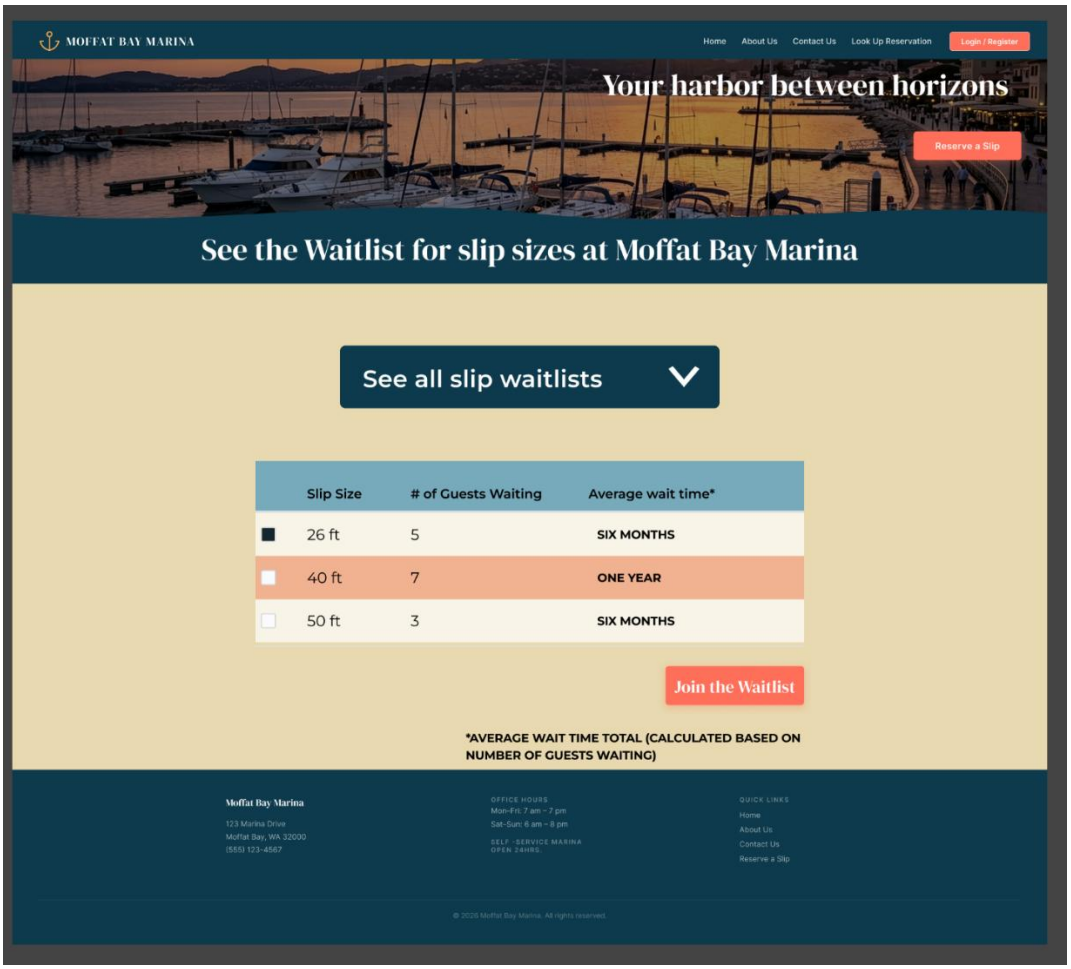


Waitlist:

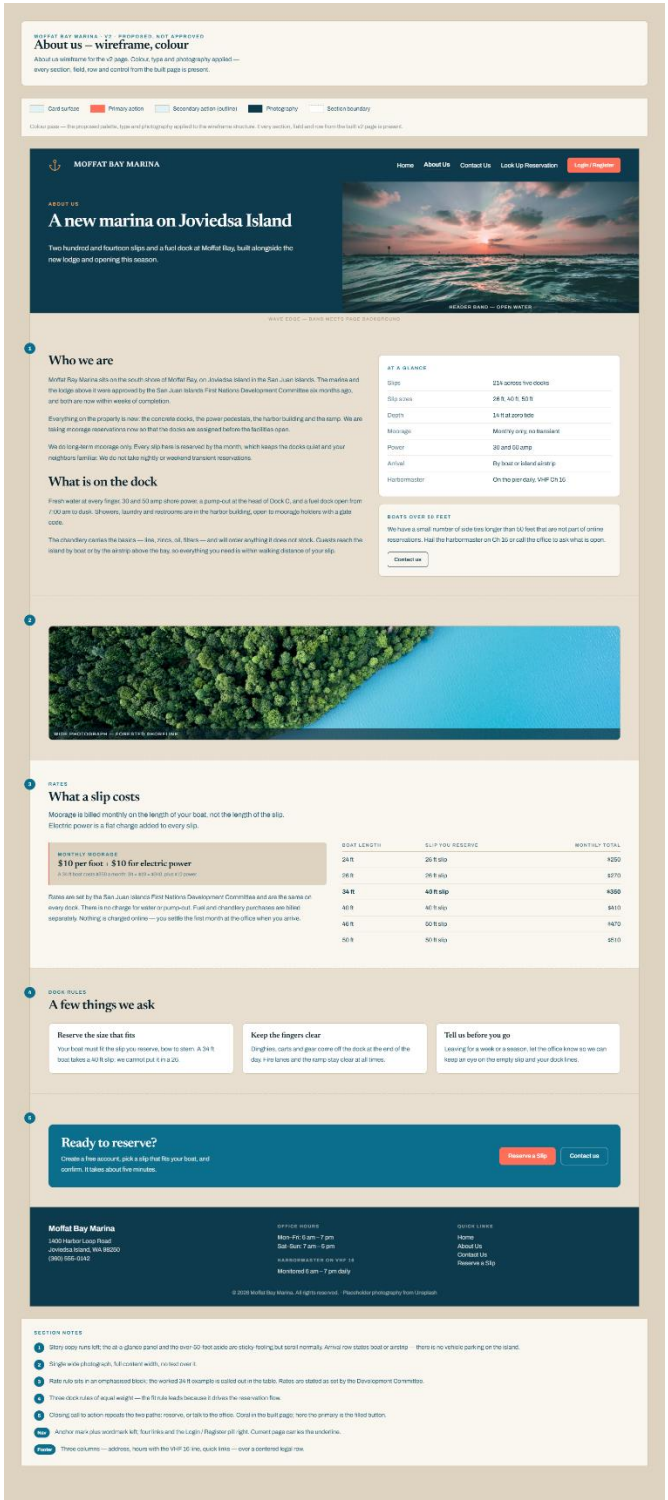

MOFFAT BAY MARINA
Home
About Us
Contact Us
Look Up Reservation
Login / Register
Your harbor between horizons
Reserve a Slip
See the Waitlist for slip sizes at Moffat Bay Marina
Select your slip size


MOFFAT BAY MARINA
Home
About Us
Contact Us
Look Up Reservation
Login / Register
Your harbor between horizons
Reserve a Slip
See the Waitlist for slip sizes at Moffat Bay Marina
Select your slip size
26 ft
40 ft
50 ft
See all slip waitlists


MOFFAT BAY MARINA
Home
About Us
Contact Us
Look Up Reservation
Login / Register
Your harbor between horizons
Reserve a Slip
See the Waitlist for slip sizes at Moffat Bay Marina
See all slip waitlists
Slip Size
of Guests Waiting
Average wait time*
26 ft
5
SIX MONTHS
40 ft
7
ONE YEAR
50 ft
3
SIX MONTHS
*AVERAGE WAIT TIME TOTAL (CALCULATED BASED ON NUMBER OF GUESTS WAITING)


MOFFAT BAY MARINA
Home
About Us
Contact Us
Look Up Reservation
Login / Register
Your harbor between horizons
Reserve a Slip
See the Waitlist for slip sizes at Moffat Bay Marina
See all slip waitlists
Slip Size
of Guests Waiting
Average wait time*
26 ft
5
SIX MONTHS
40 ft
7
ONE YEAR
50 ft
3
SIX MONTHS
Join the Waitlist
*AVERAGE WAIT TIME TOTAL (CALCULATED BASED ON NUMBER OF GUESTS WAITING)
Moffat Bay Marina
123 Marina Drive
Moffat Bay, WA 32000
(555) 123-4567
OFFICE HOURS
Mon-Fri: 7 am - 7 pm
Sat-Sun: 6 am - 8 pm
SELF - SERVICE MARINA OPEN 24HRS
QUICK LINKS
Home
About Us
Contact Us
Reserve a Slip

Wireframes:


About us – wireframe, colour
MOFFAT BAY MARINA
Home
About Us
Contact Us
Look Up Reservation
Login / Register
A new marina on Joviedsa Island
Two hundred and fourteen slips and a fuel dock at Moffat Bay, built alongside the new lodge and opening this season.
Who we are
What is on the dock
AT A GLANCE
What a slip costs
$10 per foot · $10 for electric power
A few things we ask
Reserve the size that fits
Keep the fingers clear
Tell us before you go
Ready to reserve?
Create a free account, pick a slip that fits your boat, and confirm. It takes about five minutes.
Reserve a Slip
Contact us
Moffat Bay Marina
1400 Harbor Loop Road
Joviedsa Island, WA 98250
(360) 555-0142
SECTION NOTES

MOFFAT BAY MARINA · V2 · PROPOSED, NOT APPROVED

Contact us — wireframe, colour

Contact us wireframe for the v2 page. Colour, type and photography applied — every section, field, row and control from the built page is present.

Card surface · Primary action · Secondary action (outline) · Photography · Section boundary

Colour pass — the proposed palette, type and photography applied to the wireframe structure. Every section, field and row from the built v2 page is present.

MOFFAT BAY MARINA

Home · About Us · Contact Us · Look Up Reservation · Login / Register

CONTACT US

Reach the harbor office

The office is on the pier at the head of the ramp, below the lodge. If you are on the water, hail the harbormaster before you enter the fairway.



HEADER BAND — OPEN WATER

WAVE EDGE — SAND MEETS PAGE BACKGROUND

1

Harbor office
1400 Harbor Loop Road
Joviedsa Island, WA 98250

Telephone
(360) 555-0142

Email
office@moffatbaymarina.com

Position
48°32.1'N 123°01.4'W

HARBORMASTER ON VHF

Channel 16

Hail **Moffat Bay Marina** on **VHF channel 16** for slip assignments, the fuel dock, and help on the docks. We monitor 16 from 6:00 am to 7:00 pm daily and will switch you to a working channel.

Channel 16 is also the distress and calling channel. In an emergency, call the Coast Guard first, then let us know.

OFFICE HOURS

Monday – Friday	6:00 am – 7:00 pm
Saturday	6:00 am – 7:00 pm
Sunday	7:00 am – 5:00 pm
Fuel dock	7:00 am – dusk, daily

2

Send us a message

We answer within one business day. For anything urgent, use the radio or the office line.

First name — Required
Last name — Required
Email — Required
Boat name — Optional
Length overall — 34 — in feet, bow to stern
What is this about? — Reserving a slip
Options: Reserving a slip · An existing reservation · Rates and billing · Boats over 56 feet · Fuel dock or services · Something else
Message — Tell us what you need.

Send message

3

GETTING HERE

Finding the marina

Joviedsa Island is reached by boat or by the island airstrip above the bay. By water, round the point and enter the dredged fairway between the green and red markers; the guest float is the first one inside the breakwater.



APPROACH — FORESTED SHORELINE INTO THE BAY

Moffat Bay Marina
1400 Harbor Loop Road
Joviedsa Island, WA 98250
(360) 555-0142

OFFICE HOURS
Mon–Fri: 6 am – 7 pm
Sat–Sun: 7 am – 5 pm

HARBORMASTER ON VHF 16
Monitored 6 am – 7 pm daily

QUICK LINKS
Home
About Us
Contact Us
Reserve a Slip

© 2026 Moffat Bay Marina. All rights reserved. · Placeholder photography from Unsplash

SECTION NOTES

1. Contact details as a labelled list, then the VHF block, then office hours — reading order is address, radio, hours.
2. Message form sits in a card on the right. Boat name and length overall are collected here so the office can answer moorage questions in one reply. Select options are listed under the field.
3. Getting here covers arrival by boat or airstrip — no road directions, since there is no vehicle access to the marina.

VHF — Channel 16 is the harbormaster's hailing channel, with the distress-channel caveat stated directly beneath it.

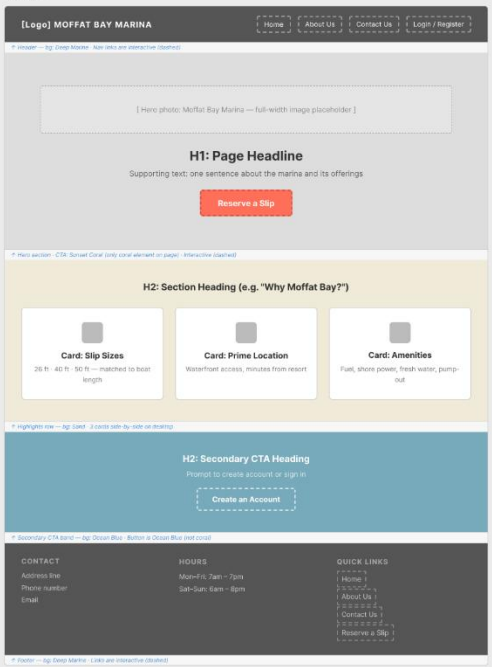
Nav — Anchor mark plus wordmark left; four links and the Login / Register pill right. Current page carries the underline.

Footer — Three columns — address, hours with the VHF 16 line, quick links — over a centered legal row.

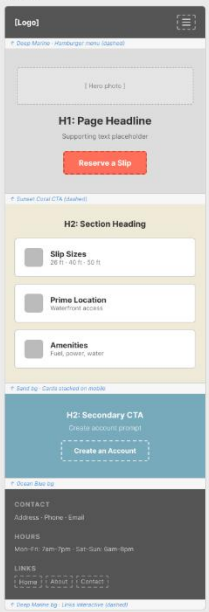
01 — Landing Page Wireframe

Moffat Bay Marina · Low-fidelity wireframe · Mobile-first

DESKTOP



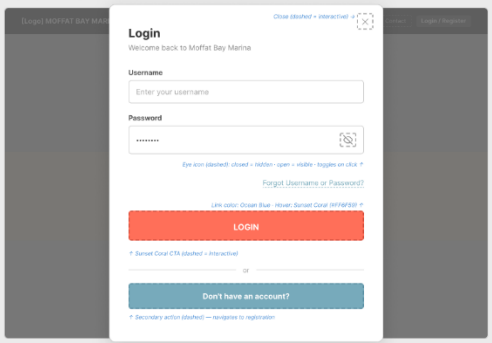
MOBILE



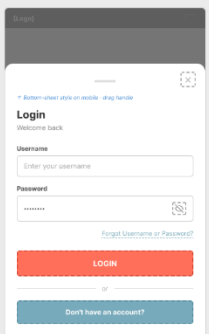
02 — Login Modal Wireframe

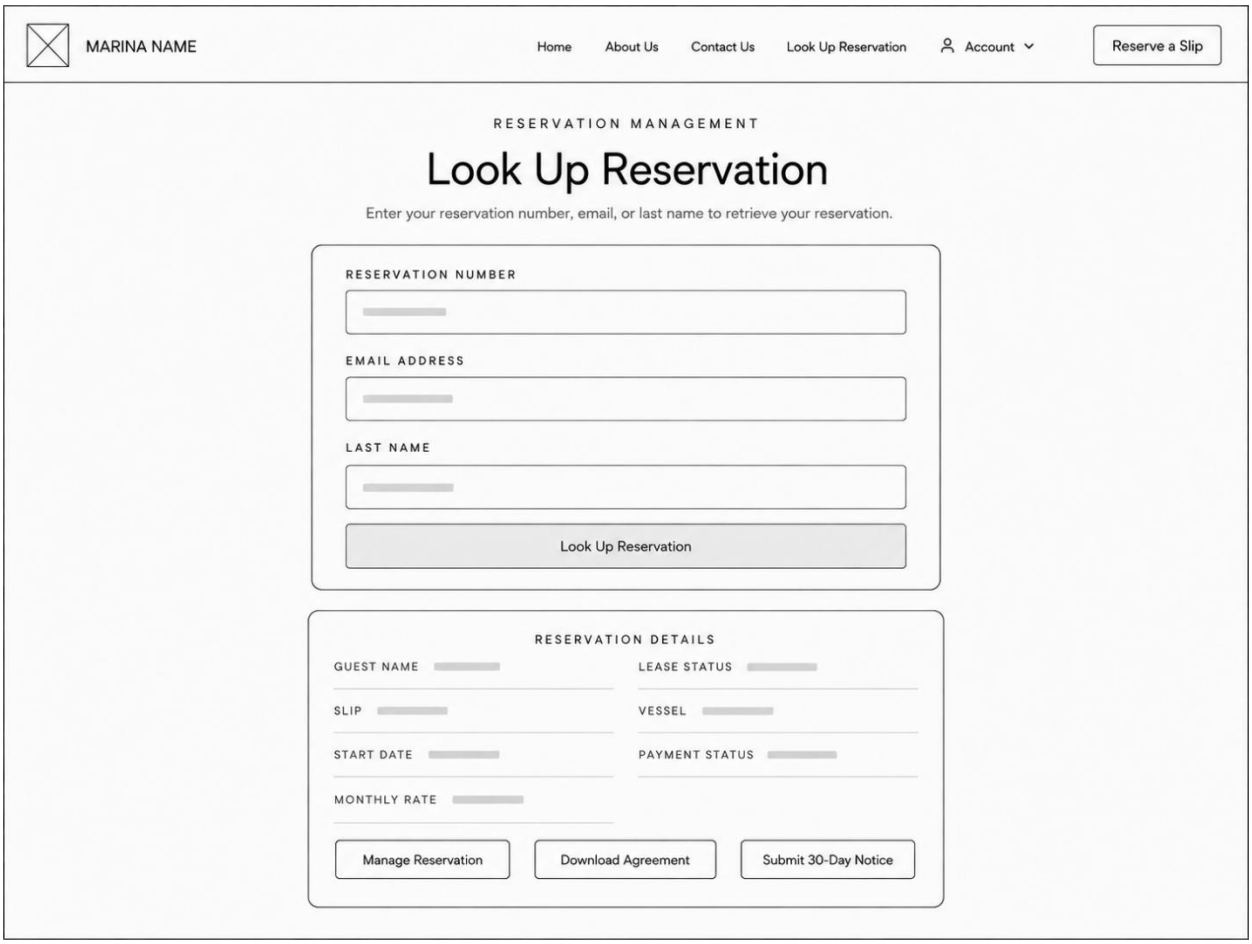
Moffat Bay Marina · Popup overlay · Mobile + Desktop

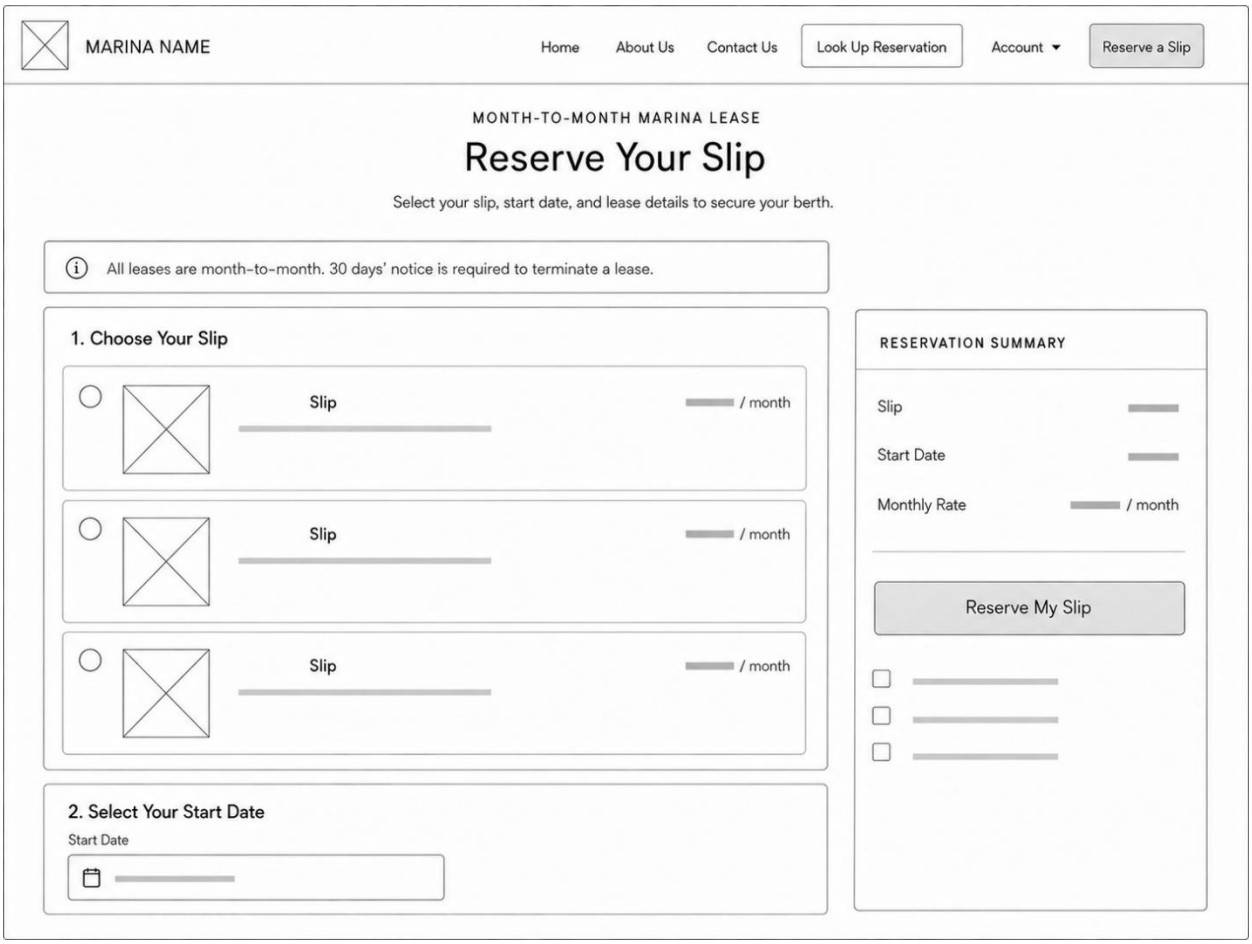
DESKTOP — 1440 PX

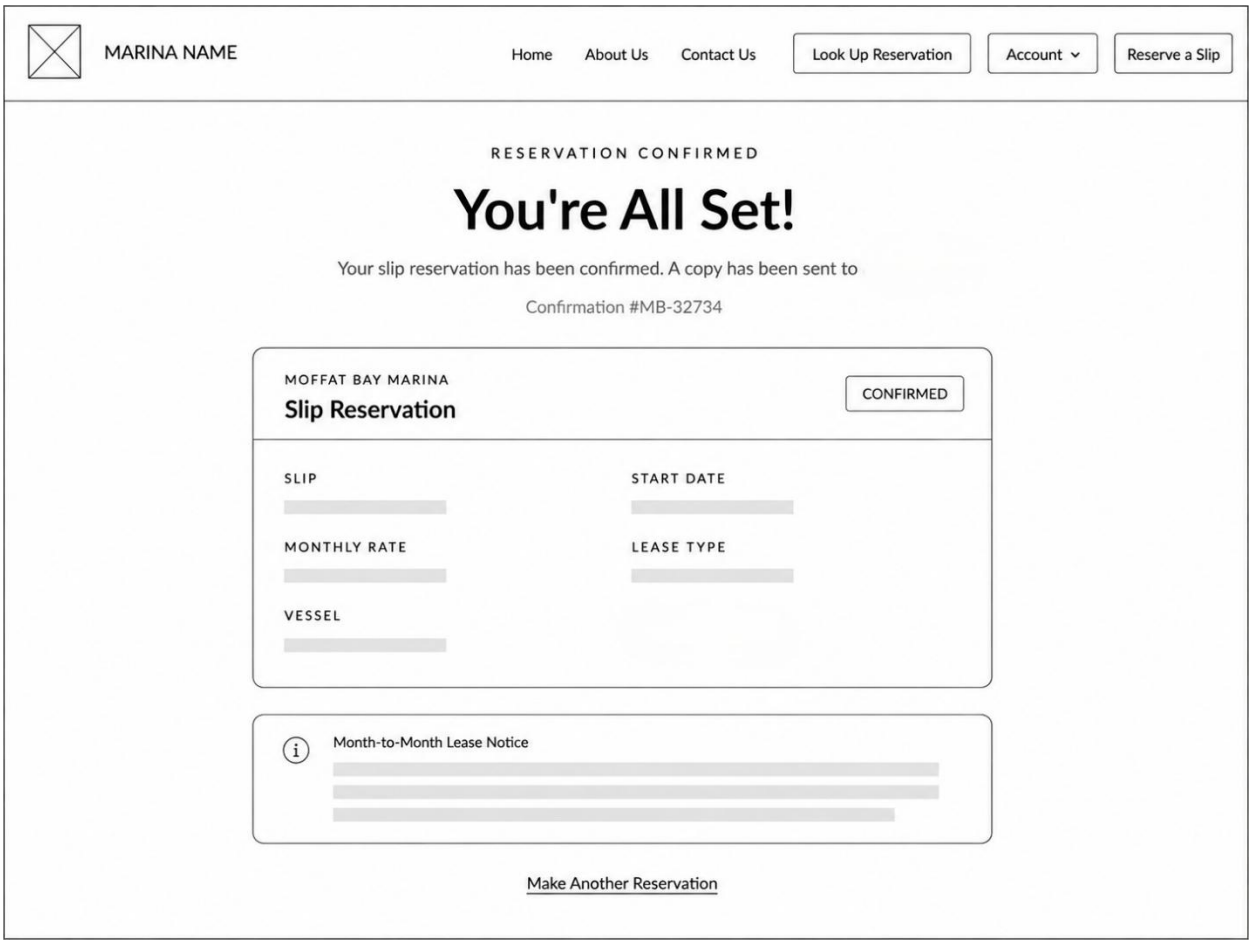


MOBILE — 375 PX

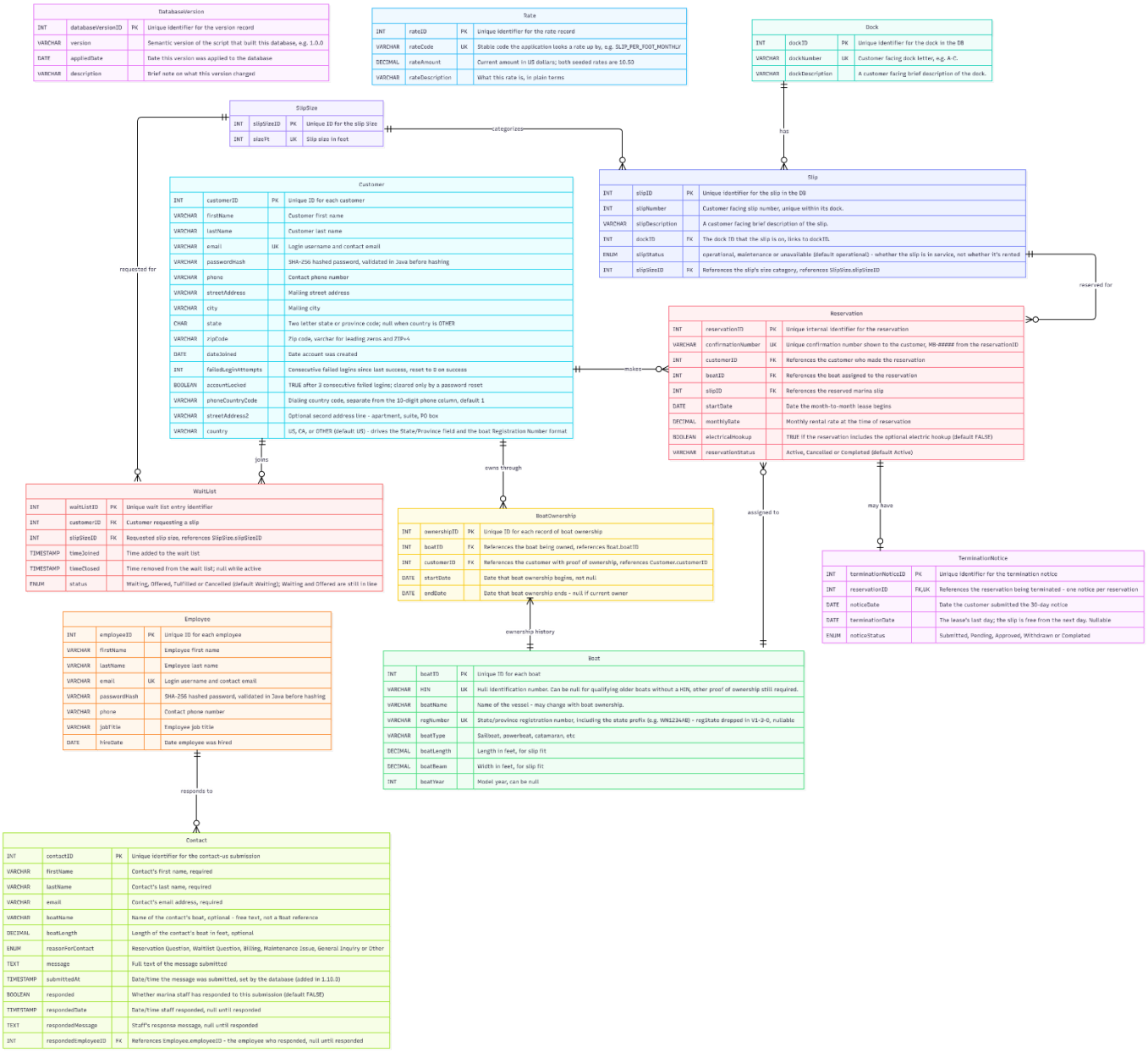



MARINA NAME
Home
About Us
Contact Us
Look Up Reservation
Account
Reserve a Slip
RESERVATION MANAGEMENT
Look Up Reservation
Enter your reservation number, email, or last name to retrieve your reservation.
RESERVATION NUMBER
EMAIL ADDRESS
LAST NAME
Look Up Reservation
RESERVATION DETAILS
GUEST NAME
LEASE STATUS
SLIP
VESSEL
START DATE
PAYMENT STATUS
MONTHLY RATE
Manage Reservation
Download Agreement
Submit 30-Day Notice


MARINA NAME
Home
About Us
Contact Us
Look Up Reservation
Account
Reserve a Slip
MONTH-TO-MONTH MARINA LEASE
Reserve Your Slip
Select your slip, start date, and lease details to secure your berth.
All leases are month-to-month. 30 days' notice is required to terminate a lease.
1. Choose Your Slip
Slip
/ month
Slip
/ month
Slip
/ month
2. Select Your Start Date
Start Date
RESERVATION SUMMARY
Slip
Start Date
Monthly Rate
/ month
Reserve My Slip


MARINA NAME
Home
About Us
Contact Us
Look Up Reservation
Account
Reserve a Slip
RESERVATION CONFIRMED
You're All Set!
Your slip reservation has been confirmed. A copy has been sent to
Confirmation #MB-32734
MOFFAT BAY MARINA
Slip Reservation
CONFIRMED
SLIP
START DATE
MONTHLY RATE
LEASE TYPE
VESSEL
Month-to-Month Lease Notice
Make Another Reservation

ERD


DatabaseVersion
INT databaseVersionID PK Unique identifier for the version record
VARCHAR version Semantic version of the script that built this database, e.g. 1.0.0
DATE appliedDate Date this version was applied to the database
VARCHAR description Brief note on what this version changed
Rate
INT rateID PK Unique identifier for the rate record
VARCHAR rateCode UK Stable code the application looks a rate up by, e.g. SLIP_PER_FOOT_MONTHLY
DECIMAL rateAmount Current amount in US dollars; both seeded rates are 10.50
VARCHAR rateDescription What this rate is, in plain terms
Dock
INT dockID PK Unique identifier for the dock in the DB
VARCHAR dockNumber UK Customer facing dock letter, e.g. A-C.
VARCHAR dockDescription A customer facing brief description of the dock.
SlipSize
INT slipSizeID PK Unique ID for the slip size
INT sizeFt UK Slip size in feet
categorizes
has
Customer
INT customerID PK Unique ID for each customer
VARCHAR firstName Customer first name
VARCHAR lastName Customer last name
VARCHAR email UK Login username and contact email
VARCHAR passwordHash SHA-256 hashed password, validated in Java before hashing
VARCHAR phone Contact phone number
VARCHAR streetAddress Mailing street address
VARCHAR city Mailing city
CHAR state Two letter state or province code; null when country is OTHER
VARCHAR zipCode Zip code, varchar for leading zeros and ZIP+4
DATE dateJoined Date account was created
INT failedLoginAttempts Consecutive failed logins since last success, reset to 0 on success
BOOLEAN accountLocked TRUE after 3 consecutive failed logins; cleared only by a password reset
VARCHAR phoneCountryCode Dialing country code, separate from the 10-digit phone column, default 1
VARCHAR streetAddress2 Optional second address line - apartment, suite, PO box
VARCHAR country US, CA, or OTHER (default US) - drives the State/Province field and the boat Registration Number format
Slip
INT slipID PK Unique identifier for the slip in the DB
INT slipNumber Customer facing slip number, unique within its dock.
VARCHAR slipDescription A customer facing brief description of the slip.
INT dockID FK The dock ID that the slip is on, links to dockTbl.
ENUM slipStatus operational, maintenance or unavailable (default operational) - whether the slip is in service, not whether it's rented
INT slipSizeID FK References the slip's size category, references SlipSize.slipSizeID
requested for
reserved for
makes
Reservation
INT reservationID PK Unique internal identifier for the reservation
VARCHAR confirmationNumber UK Unique confirmation number shown to the customer, MB-##### from the reservationID
INT customerID FK References the customer who made the reservation
INT boatID FK References the boat assigned to the reservation
INT slipID FK References the reserved marina slip
DATE startDate Date the month-to-month lease begins
DECIMAL monthlyRate Monthly rental rate at the time of reservation
BOOLEAN electricalHookup TRUE if the reservation includes the optional electric hookup (default FALSE)
VARCHAR reservationStatus Active, Cancelled or Completed (default Active)
joins
owns through
assigned to
may have
WaitList
INT waitListID PK Unique wait list entry identifier
INT customerID FK Customer requesting a slip
INT slipSizeID FK Requested slip size, references SlipSize.slipSizeID
TIMESTAMP timeJoined Time added to the wait list
TIMESTAMP timeClosed Time removed from the wait list; null while active
ENUM status Waiting, Offered, Fulfilled or Cancelled (default Waiting); Waiting and Offered are still in line
BoatOwnership
INT ownershipID PK Unique ID for each record of boat ownership
INT boatID FK References the boat being owned, references Boat.boatID
INT customerID FK References the customer with proof of ownership, references Customer.customerID
DATE startDate Date that boat ownership begins, not null
DATE endDate Date that boat ownership ends - null if current owner
TerminationNotice
INT terminationNoticeID PK Unique identifier for the termination notice
INT reservationID FK,UK References the reservation being terminated - one notice per reservation
DATE noticeDate Date the customer submitted the 30-day notice
DATE terminationDate The lease's last day; the slip is free from the next day. Nullable
ENUM noticeStatus Submitted, Pending, Approved, Withdrawn or Completed
ownership history
Employee
INT employeeID PK Unique ID for each employee
VARCHAR firstName Employee first name
VARCHAR lastName Employee last name
VARCHAR email UK Login username and contact email
VARCHAR passwordHash SHA-256 hashed password, validated in Java before hashing
VARCHAR phone Contact phone number
VARCHAR jobTitle Employee job title
DATE hireDate Date employee was hired
Boat
INT boatID PK Unique ID for each boat
VARCHAR HIN UK Hull Identification number. Can be null for qualifying older boats without a HIN, other proof of ownership still required.
VARCHAR boatName Name of the vessel - may change with boat ownership.
VARCHAR regNumber UK State/province registration number, including the state prefix (e.g. WN1234AB) - regState dropped in V1-2-0, nullable
VARCHAR boatType Sailboat, powerboat, catamaran, etc
DECIMAL boatLength Length in feet, for slip fit
DECIMAL boatBeam Width in feet, for slip fit
INT boatYear Model year, can be null
responds to
Contact
INT contactID PK Unique identifier for the contact-us submission
VARCHAR firstName Contact's first name, required
VARCHAR lastName Contact's last name, required
VARCHAR email Contact's email address, required
VARCHAR boatName Name of the contact's boat, optional - free text, not a Boat reference
DECIMAL boatLength Length of the contact's boat in feet, optional
ENUM reasonForContact Reservation Question, Waitlist Question, Billing, Maintenance Issue, General Inquiry or Other
TEXT message Full text of the message submitted
TIMESTAMP submittedAt Date/time the message was submitted, set by the database (added in 1.10.0)
BOOLEAN responded Whether marina staff has responded to this submission (default FALSE)
TIMESTAMP respondedDate Date/time staff responded, null until responded
TEXT respondedMessage Staff's response message, null until responded
INT respondedEmployeeID FK References Employee.employeeID - the employee who responded, null until responded

Moffat Bay Marina - Group Task List & Work Estimations

Each page is broken into Front End, Back End, and Database needs. Estimated hours are approximate - see citations below.

Task	Description	Estimated Hours	Due Date
MODULE 1 - Project Setup & Requirements Gathering (due Aug 16) - DONE			
Requirements Gathering	Draft data requirements, user personas, and user stories for the TDD	6 hrs	Aug 16 (Done)
Project Setup	Confirm Marina project, set up GitHub repo, shared MySQL environment, comms channel	4 hrs	Aug 16 (Done)
MODULE 2 - Prototype (due Aug 23)			
Page Prototypes	Functional prototype for all 9 pages (Landing, About Us, Contact Us, Wait List Lookup, Registration, Login, Slip Reservation, Reservation Summary, Look Up Reservation)	9 hrs	Aug 23
Landing Page Mockup	Realistic mockup (color, layout, font) of the Landing page	3 hrs	Aug 23
MODULE 3 - ERD (due Aug 30) - DATABASE			
Entity Relationship Diagram	Design the ERD covering customers, slips, reservations, and wait list, and how they relate	6 hrs	Aug 30

Task	Description	Estimated Hours	Due Date
MODULE 4 - Database Development, Sprint 1 (due Aug 30) - DATABASE			
Database Schema	Build the MySQL schema from the ERD (customers, slips, reservations, wait list tables)	6 hrs	Aug 30
Slip Inventory	Load slip inventory by size category (26 ft, 40 ft, 50 ft) based on the marina map	2 hrs	Aug 30
Shared Data Access Functions	Build shared functions so every page has a consistent way to read/write the database	7 hrs	Aug 30
Password Hashing	Research and implement password hashing approach (cited source)	4 hrs	Aug 30
MODULE 5 - Web Dev 1, Sprint 2 Begins (due Sep 6)			
Landing Page - Front End	HTML/CSS build	3 hrs	Sep 6
Landing Page - Back End	Any dynamic content	2 hrs	Sep 6
Landing Page - Database	*None - static content*	-	*Sep 6*

Task	Description	Estimated Hours	Due Date
Login Page - Front End	Username/password form	2 hrs	Sep 6
Login Page - Back End	Check credentials, hash check, start session	4 hrs	Sep 6
Login Page - Database	*Reads customers table (username, password hash)*	-	*Sep 6*
Registration Page - Front End	Registration form	2 hrs	Sep 6
Registration Page - Back End	Create customer record, validate email format, hash password	4 hrs	Sep 6
Registration Page - Database	*Writes to customers table (new row, unique CustomerID, hashed password)*	-	*Sep 6*
Module 5 Test Plan	6 functional tests (2 per page) using TestPlanTemplate.docx, run and document with screenshots	3 hrs	Sep 6
MODULES 6 & 7 - Web Dev 2 & 3, Sprint 2 Continues (due Sep 13)			
Slip Reservation Page - Front End	Reservation form	3 hrs	Sep 13
Slip Reservation Page - Back End	Boat-length-to-slip-size matching, availability check	5 hrs	Sep 13

Task	Description	Estimated Hours	Due Date
Slip Reservation Page - Database	*Reads slips (availability by size), writes reservations*	-	*Sep 13*
About Us Page - Front End	HTML/CSS build	2 hrs	Sep 13
About Us Page - Back End	Any dynamic content	1 hr	Sep 13
About Us Page - Database	*None - static content*	-	*Sep 13*
Reservation Summary Page - Front End	Summary display	2 hrs	Sep 13
Reservation Summary Page - Back End	Pull reservation + slip details for display	3 hrs	Sep 13
Reservation Summary Page - Database	*Reads reservations, slips, customers*	-	*Sep 13*
Module 7 Test Plan	4 functional tests (2 per page)	2 hrs	Sep 13
Scrum 2 Discussion	Initial post plus 3 peer responses (per person)	1 hr/person	Sep 13
MODULE 8 - Web Dev 4, Sprint 3 Begins (due Sep 20)			

Task	Description	Estimated Hours	Due Date
Contact Us Page - Front End	HTML/CSS build	2 hrs	Sep 20
Contact Us Page - Back End	Any dynamic content	1 hr	Sep 20
Contact Us Page - Database	*None - static content (unless a stored contact form is added later)*	-	*Sep 20*
Look Up Reservation Page - Front End	Lookup form	2 hrs	Sep 20
Look Up Reservation Page - Back End	Query reservation by ID/customer	3 hrs	Sep 20
Look Up Reservation Page - Database	*Reads reservations*	-	*Sep 20*
Module 8 Test Plan	4 functional tests (2 per page)	2 hrs	Sep 20
Scrum 3 Discussion	Initial post plus 3 peer responses (per person)	1 hr/person	Sep 20
MODULE 9 - Web Dev 5, Sprint 3 Wraps Up (due Sep 27)			
Wait List Lookup Page - Front End	Wait list display	2 hrs	Sep 27

Task	Description	Estimated Hours	Due Date
Wait List Lookup Page - Back End	Wait-list logic when a slip size is full	4 hrs	Sep 27
Wait List Lookup Page - Database	*Reads/writes wait_list, reads slips*	-	*Sep 27*
Module 9 Test Plan	2 functional tests	1 hr	Sep 27
MODULE 10 - QA & Peer Review (due Oct 4)			
QA Presentation	Team intro, prototypes, ERD, 5+ tests and results, lessons learned	4 hrs	Oct 4
Screen Capture Walkthrough	Narrated recording walking through every page of the site	3 hrs	Oct 4
MODULE 11 - QA Testing & Delivery (due Oct 11)			
Peer Critiques	2 critiques of other teams, 150+ words each, 2 strengths/2 improvements	2 hrs/person	Oct 11
Peer Feedback Fixes	Fix issues raised in peer feedback across pages/database	5-8 hrs	Oct 11
Changes Write-Up	Document the changes made based on feedback	2 hrs	Oct 11
Final Submission	Zip and submit the final project	1 hr	Oct 11

How These Estimates Were Generated

We started with two numbers we found while researching typical web project timelines: about 40 hours for planning and design, and roughly 5 hours to build each page. For the login page, we used a starting point of 1 to 3 hours for a basic username and password check. From there, we adjusted each task up or down by hand. A page like Slip Reservation, which has to match a boat's length to an available slip, took more time than something like the About Us page. It's worth knowing that both of these numbers came from articles written to help price out client projects, not from a formal study of how long development actually takes. So, think of them as a solid starting point, not a guarantee.

References

Ceccarelli, G. (2026, August 1). Technical design documents: A practical guide. Stoa. https://withstoa.com/blog/technical-design-documents

City of Monterey. (n.d.). Harbor and marina staff directory [Photograph]. https://monterey.gov/your_city_hall/departments/public_works/harbor_and_marina/staff_directory/index.php#prettyPhoto

Discover Boating. (n.d.). Boating glossary. https://www.discoverboating.com/glossary-items

Erickson, J. (2024, August 29). MySQL: Understanding what it is and how it's used. Oracle. https://www.oracle.com/mysql/what-is-mysql/

Givnish, C. (2015, July 21). User-centered design: User personas vs. user stories. Function1. https://www.function1.com/2015/07/user-centered-design-user-personas-vs-user-stories

Google. (2026). Portrait of marina customer [AI-generated image]. Gemini.

Martin, M. (2025, February 12). How much does it cost to develop a website? DesignRush. https://www.designrush.com/agency/web-development-companies/trends/cost-of-web-development

MDN contributors. (2025a, July 11). Glossary of web terms. MDN Web Docs. https://developer.mozilla.org/en-US/docs/Glossary

MDN contributors. (2025b, November 7). JavaScript Guide. MDN Web Docs. https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide

MDN contributors. (2025c, December 22). HTML: HyperText Markup Language. MDN Web Docs. https://developer.mozilla.org/en-US/docs/Web/HTML

NS804. (2019, June 4). How much does it cost to implement login capabilities? https://www.ns804.com/blog/how-much-does-it-cost-to-implement-login-capabilities/

Oracle. (n.d.). MySQL glossary. In MySQL 8.4 reference manual. https://dev.mysql.com/doc/refman/8.4/en/glossary.html